

Question:

I have a bag with yellow and blue marbles in it. At the moment, the ratio of blue marbles to yellow marbles is 4:3. If I add 5 blue marbles and remove 3 yellow marbles, the ratio will be 7:3. How many blue marbles were in the bag before I added more?

Response:

<think>Let the initial number of blue marbles be $4x$ and yellow marbles be $3x$. After adding 5 blue marbles and removing 3 yellow marbles, the new ratio is 7:3. So, we **get** $3(4x+5) = 7(3x-3) \dots x=4$. The blue count is 16.

